

Trigonometric Angle Identities

TRIGONOMETRIC ANGLE IDENTITIES (PRINTABLE CHEAT-SHEET)

1. Co-function identities (Complementary angles)

$$\sin(\pi/2 - x) = \cos x$$

$$\cos(\pi/2 - x) = \sin x$$

$$\tan(\pi/2 - x) = \cot x$$

$$\cot(\pi/2 - x) = \tan x$$

$$\sec(\pi/2 - x) = \csc x$$

$$\csc(\pi/2 - x) = \sec x$$

2. Supplementary angles ($\pi - x$)

$$\sin(\pi - x) = \sin x$$

$$\cos(\pi - x) = -\cos x$$

$$\tan(\pi - x) = -\tan x$$

$$\cot(\pi - x) = -\cot x$$

3. Negative angles ($-x$)

$$\sin(-x) = -\sin x$$

$$\cos(-x) = \cos x$$

$$\tan(-x) = -\tan x$$

$$\cot(-x) = -\cot x$$

$$\sec(-x) = \sec x$$

$$\csc(-x) = -\csc x$$

4. $\pi + x$ ($180^\circ + x$)

$$\sin(\pi + x) = -\sin x$$

$$\cos(\pi + x) = -\cos x$$

$$\tan(\pi + x) = \tan x$$

$$\cot(\pi + x) = \cot x$$

5. $2\pi - x$ ($360^\circ - x$)

$$\sin(2\pi - x) = -\sin x$$

$$\cos(2\pi - x) = \cos x$$

$$\tan(2\pi - x) = -\tan x$$

$$\cot(2\pi - x) = -\cot x$$

6. Extra

$$\sin(\pi/2 + x) = \cos x$$

$$\cos(\pi/2 + x) = -\sin x$$

$$\tan(\pi/2 + x) = -\cot x$$